Forest Height Mapping With Multifrequency SAR in Mediterranean Forests

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***Abstract*— The importance of mapping the forest height (FH) is increasing due to the more frequent impacts of climate change in the society (wildfires, droughts, and extreme weather events). Remote Sensing is often used for mapping this variable; however, it usually relies in costly and extensive field or airborne campaigns. In addition, when using synthetic aperture radar (SAR), most approaches do not use freely available data. Considering this, in this work a model is proposed that resorts to Advanced Land Observing Satellite 2 (ALOS-2), Sentinel-1 (S1), and ancillary data. Airborne laser scanning (ALS) data are used for local calibration but, with the aim of developing a more scalable model, the latter is optimized to work with small calibration datasets (representative of just 25% of the study area to be mapped). With this purpose, the model combines a featuring generation and a features' processing stage with a stacking regressor to produce estimates at the pixel level. Their impact was assessed, and an improvement of 8.11 and 2.01 pp in the relative root mean square error (rRMSE) was achieved by including the features' generation and features' processing stages, respectively. In addition, when the multifrequency dataset was used, the model achieved an rRMSE better than when using only a C-band dataset (S1) or only an L-band dataset (ALOS-2), respectively, by 4.21 and 3.05 pp. Finally, the model achieved an average R^2/rRMSE of 0.6240%/24.30% and 0.5901%/22.64% for the validation and test study areas, respectively. The proposed approach revealed to be effective on mapping the FH resorting to multifrequency SAR and small calibration datasets acquired by ALS.**

***Index Terms*— Advanced Land Observing Satellite 2 (ALOS-2), backscatter derivatives, climate change, forest height (FH), gradient boosting, multifrequency synthetic aperture radar (SAR), random forest, regression, remote sensing, Sentinel-1 (S1), stacking regressor, synergy.**

I. Introduction

FOREST height (FH) mapping is becoming increasingly important due to climate change, which leads to a greater need for forest monitoring. The information provided by FH maps is not limited to the height itself, as it can be used as proxy of other variables as the aboveground biomass [1], biodiversity [2], carbon cycle [3], and others [4]. Therefore, these maps are important tools for monitoring and reporting the implementation of sustainable forest management practices [5]. In addition, the knowledge of FH is helpful in the context of wildfires, in monitoring forests before the fire season [6]. Sections I-A and I-B present the state-of-the-art on FH mapping and the research proposal of this article.

A. State-of-the-Art

In the literature, light detection and ranging (LiDAR), synthetic aperture radar (SAR), and multispectral imaging [7], [8], [9] are used to estimate the FH. Airborne laser scanning (ALS) campaigns are the most reliable method in remote sensing to measure the forest inventory variables [10], [11], [12]. The usual way to retrieve the FH from ALS is to generate a canopy height model (CHM) from the point cloud. Despite their accuracy, ALS is expensive and does not have worldwide coverage. Spaceborne LiDAR sensors, such as global ecosystem dynamics investigation (GEDI) and ice, cloud, and land elevation satellite-2 (ICESat-2) [13], [14], overcome the issues, but their spatial distribution is limited, and the measurements obtained are less accurate.

One of the main alternatives to ALS is the use of SAR sensors. When using SAR, three approaches can be followed for FH retrieval: interferometric SAR (InSAR), tomographic SAR (TomoSAR), and the use of backscatter directly. When resorting to InSAR, models based on the relationship between the coherence and FH, as the random volume over ground (RVoG) model [15], [16] or the SINC model [17], [18], are commonly used. When adopting a TomoSAR approach, the FH is retrieved from the tomograms derived from the acquired data [19], [20]. Since TomoSAR mostly uses airborne sensors, it has the same drawbacks as the ALS campaigns. Alternatively, some authors used the backscatter directly to estimate the FH. In [21], an exponential regression was proposed for mapping FH. Machine learning regressors, such as random forests [22] and artificial neural networks [23], have also been used as regressors of the FH. Comparing to parametric regressors, machine learning regressors are more capable of modeling nonlinear relationships, make fewer assumptions, and are more scalable [24]. Finally, in [3], a CHM was generated resorting to multifrequency SAR, using the X-band for the surface model and the L-band for the terrain model, being the CHM the difference between both.

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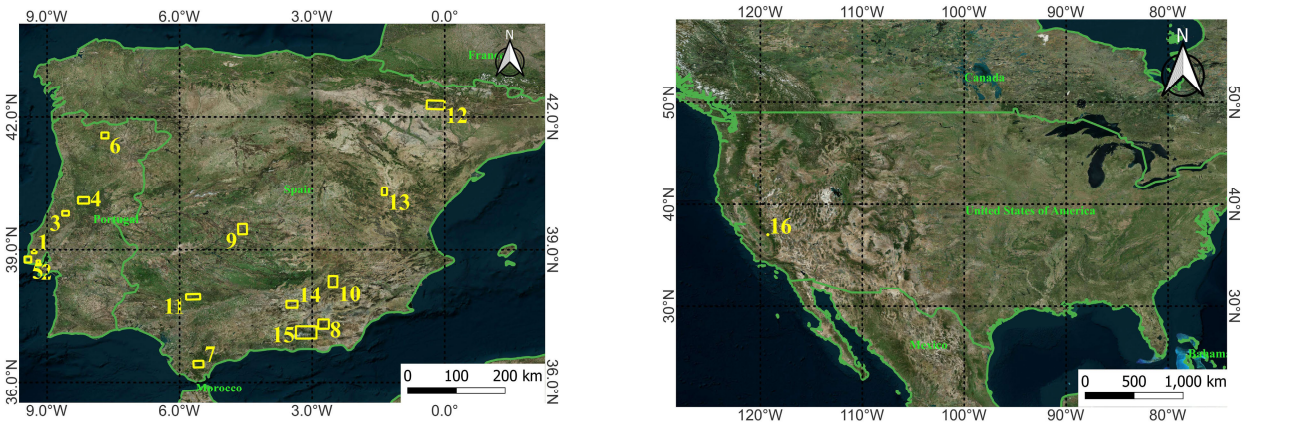
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(a) (b)

Fig. 1. Study areas' location. (a) Portugal and Spain. (b) California. (The study areas' ids are provided in Table I.)

The combination of LiDAR and SAR sensors for mapping the FH can also be found in the current state-of-art. The synergy is used to support the inversion of the RVoG and SINC models [25], [26]. Alternatively, some authors used the LiDAR data as reference fitting regressors that resorted to SAR features [27].

The InSAR approach is effective, but it usually relies on nonfreely available data (e.g., TanDEM-X), increasing the costs of using it in an operational scenario. More research on the integration of LiDAR and SAR will be beneficial, also in consideration of the upcoming NASA ISRO SAR Mission (NISAR).

B. Research Proposal

Considering the analysis of the state-of-the-art, this article proposes a locally calibrated model capable of mapping the FH for study areas with different types of land cover using multifrequency SAR data from the Advanced Land Observing Satellite 2 (ALOS-2) and Sentinel-1 (S1). The model was optimized to work with small local calibration datasets (acquired through ALS), representative of a maximum coverage of 25% of the study areas being mapped (tests were carried out using even smaller calibration datasets). This will enable the mapping of wider areas with smaller ALS campaigns (or smaller field campaigns) thus enabling an application of forest survey even where calibration data are scarcer. The model used stacking regression, and it was tested in several Mediterranean study areas (from Portugal, Spain, and California). In addition, the accuracy impact of using a multifrequency dataset that includes the L-band (ALOS-2) and C-band (S1) was also assessed. This article aims to answer the following research questions.

1) Is it possible to map the FH using multifrequency SAR data and a stacking regressor, with small calibration datasets?
2) Which is the impact of using multifrequency SAR for FH mapping?

This article is composed of six sections. After the initial introduction (see Section I), Sections II and III describe the study areas, and the data and tools used for the implementation and validation of the proposed model. In Section IV, an in-depth presentation of the model is done, and the validation and testing procedures are explained. Then, Sections V and VI present and discuss the results. Section VI contains also a subsection comparing the results in this article with relevant works found in the literature. The article ends with Section VII where the main conclusions are drawn, and future developments are discussed.

II. Study Areas

Although many FH mapping techniques were proposed in the literature, only a few were applied to Mediterranean regions [9], [18], [28]. For this study, a set of 16 areas across Portugal, Spain, and California were selected. The selection of these areas was limited by the availability of ALS campaigns. In the Spanish case, as there was countrywide ALS data available, nine study areas from different parts of the country were selected. Fig. 1 shows the geographic location of the 16 identified study areas.

Regarding forest composition of the study areas, Table I shows that four of them are dominated by broad-leaved trees, five by coniferous, and the remaining six are mixed tree species. Since the land cover land use (LCLU) for Soaproot Saddle did not classify forests into broad-leaved and coniferous, this study area was not considered in the previous grouping. It should be noted that different LCLU maps were used for Portugal, Spain, and California (described in Section III-D). Another important detail was that the Spanish study areas and the Monsanto area were located inside natural parks.

III. Data and Tools

The model here presented makes use of multifrequency SAR data from ALOS-2 and S1, LiDAR data from different ALS campaigns, and ancillary data including digital elevation models (DEMs) and LCLU maps.

A. Airborne Laser Scanning

The ALS data were acquired in three different campaigns, depending on the country. As the campaigns did not use

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TABLE I
STUDY AREAS' FOREST COMPOSITION ACCORDING TO LCLU MAP

	Study area (ID)	Forest Type (%)		
		Broad-leaved	Conifer-ous	Mixed
Portugal	Mafra (1)	85	15	-
	Monsanto (2)	17	83	-
	Pombal (3)	69	31	-
	Serra Lousã (4)	44	56	-
	Sintra-Cascais (5)	34	66	-
	VPA (6)	17	83	-
Spain	Alcornacles (7)	93	4	3
	Baza (8)	2	91	7
	Cabaneros (9)	90	4	6
	Cazorla (10)	5	84	11
	Constantina (11)	~100	0	~0
	Guara (12)	51	21	28
	Rodeno (13)	1	90	9
	Sierra Magina (14)	40	44	16
	Sierra Nevada (15)	20	71	9
U.S.	Soaproot Saddle (16)	Evergreen (~100)		

the same sensor (despite these all being LiDAR discrete return sensors), point clouds with different specifications were obtained.

The Portuguese ALS campaign was carried out within the scope of the ÁGIL project [29], using data from six flights, each covering one study area. All of them used the Teledyne Galaxy PRIME Airborne Lidar Terrain Mapper with Swath TRAK sensor, which generated point clouds with point densities between 10.02 and 13.88 points/m^2. The six flights took place in June 2020.

The National Plan for Aerial Orthophotography (PNOA) [30], started in 2009, covered the entire Spanish territory in six-year periods. Currently, it is on its third coverage (started in 2023). In this work, only data from the second coverage (occurred between 2015 and 2021) were used. To obtain the point clouds for the nine Spanish study areas, data from seven different flights were needed. Leica ALS80, Optech Prime, Optech T2000, and Riegl VQ-1560i were the sensors used in these flights. The resulting point densities ranged from 1.00 to 2.00 points/m^2.

Study Area 16 is a plot in Sierra National Forest, California, established by the National Ecological Observatory Network (NEON) project [31] in 2015. ALS campaigns covering this plot were carried out in 2013, 2017, 2018, 2019, and 2021. In this work, data from the 2019 flight were used, which had the ALTM Gemini LiDAR as sensor. A point density from 2.00 to 6.00 points/m^2 was achieved.

B. Advanced Land Observing Satellite 2

Global L-band (1.200 GHz) mosaic provided annually by the ALOS Research and Application Project was considered. The generation of this mosaic relied on data acquired in fine beam dual mode (a stripmap mode) and, whenever gap-filling was needed, in wide swath beam dual mode (a ScanSAR mode), for the HH + HV dual-polarization. The product was already orthorectified and radiometrically corrected. The annual observation corresponded to the linear amplitude of the terrain-flattened gamma nought values in the polarizations HH and HV, at a spatial resolution of 25 m. The full description of the product is provided in [32]. For each study area, an ALOS-2 observation of the same year as the ALS campaign was used.

C. Sentinel-1

A wide diversity of S1 products (acquired in the C-band, 5.405 GHz) can be downloaded from the Copernicus Open Access Hub (https://scihub.copernicus.eu/dhus/). However, only those guaranteeing continuous and worldwide coverage meet the criteria for operational scenarios. In view of this, the ground range detected (GRD) products acquired in interferometric wide, in VV + VH dual-polarization, were used in this study. The GRD is a level 1 product providing backscatter measurements with a spatial resolution of 10 m and a temporal resolution of six days. For each study area, the S1 observations spanned a six-month period (giving a total of 29–31 observations per study area). The year of these observations was aligned with the corresponding ALS campaign. When compared with the L-band of ALOS-2, the higher frequency of S1 leads to lower penetration depth in the forest canopy.

D. Ancillary Data

In addition to the LiDAR and SAR data, DEM, LCLU maps, and the ALOS-2 and S1 local incidence angles were included in the model. The first two were used in the preprocessing stage and as features for FH regression.

The DEM was used in both the terrain-flattening and terrain-correction steps of the S1 products' preprocessing. Two DEMs were tested to compare performances, the one acquired by the Shuttle Radar Topography Mission (SRTM) [33] and the Digital Surface Model provided by JAXA, ALOS World 3-D (AW3D30, here used as DEM) [34].

Three different LCLU maps were used in the scope of this article. For Portugal, it was the Land Cover Map 2018 (COS 2018, Land Cover Map) [35], which is a map produced for the entire country. It has a total of 83 classes, ten of

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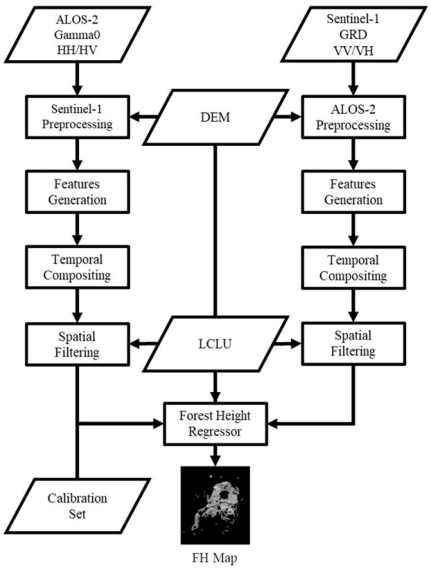


Fig. 2. Model overview.

which are related to forests. These classes can be divided into broad-leaved and coniferous species. COS 2018 has a minimum mapping unit of 1 ha. The Corine Land Cover 2018 is an LCLU map produced for Europe [36]. Among its 44 different classes, three are dedicated to forests. The minimum mapping unit of this map is 25 ha. For California, the National Land Cover Database (NLCD) 2019 was used [37]. It provides 20 classification classes with three forest classes for the territory of the United States. It has a spatial resolution of 30 m.

The inclusion of the local incidence angles (from ALOS-2 and S1) aimed to provide the regressor with more information about the acquisition, in the attempt of improving the quality of its estimates. A summary of the used data per study area is displayed in Table II.

IV. FH Mapping Model

The proposed modeling process for FH mapping can be divided into five stages: SAR preprocessing, features generation, temporal compositing, spatial filtering, and FH mapping (an overview of the model is shown in Fig. 2). For each stage, different methods were evaluated, to understand the most suitable for FH retrieval. In addition to the presentation of these stages, this section also addresses the generation of the CHM from the ALS point clouds and the model building and testing.

A. SAR Preprocessing

The preprocessing of ALOS-2 gamma nought data consisted of two steps: speckle filtering and conversion to decibels (dB). The dB conversion of the digital number was done resorting to the following equation:

$$\gamma = 10\log_{10}(\mathrm{DN}) - 83.0. \quad (1)$$

From the S1 GRD products, the terrain-flattened sigma nought and gamma nought backscatter coefficients (BCs) were computed. The preprocessing started by updating the orbit files and by subsetting the study area from the observation. Next, the thermal noise was removed, reducing the inter subswath discontinuities of the backscatter signal. Finally, the sigma and gamma nought were obtained by the radiometric terrain-flattening operator, which compensated for the terrain effects, especially over complex areas. For this step, which required a DEM, the use of the SRTM and AW3D30 was compared. Then, the BCs were speckle-filtered, and terrain-corrected. The latter resorted to the range Doppler terrain correction operator (which also resorted to a DEM), which decreased the effects geometrical distortions and resampled the observations to a spatial resolution of 25 m, matching the native resolution of the ALOS-2 data. The resampling was done with bicubic interpolation. The S1 preprocessing ends with dB conversion of BCs. A block diagram of the S1 preprocessing stage is shown in Fig. 3.

Three speckle filters were compared: Lee, Lee sigma, and refined Lee, which use different strategies to avoid removing linear features from the images [38] (that may also be present in forests). Each filter was tested with different configurations, as follows:

1) for the Lee filter three different kernels: 3×3, 5×5, 7×7;
2) for the Lee sigma two different kernels, 5×5 and 7×7, and three different sigma values [0.5, 0.7, 0.9];
3) for the refined Lee only the 7×7 kernel.

Therefore, ten different speckle filter scenarios were tested. Note that the kernel sizes did not exceed 7×7, to avoid too much smoothing.

At the end of this stage, the L-band gamma nought for HH and HV (ALOS-2) and the C-band gamma and sigma nought for VV and VH (S1) were generated. The preprocessing was done using the Sentinel Application Platform (SNAP) software.

B. Features Generation

A total of 72 features (24 for ALOS-2 and 48 for S1) were generated from the BCs. These can be grouped into three groups: 1) textural; 2) polarization ratios and normalized differences; and 3) principal components.

The textural features were computed through the gray-level co-occurrence matrix (GLCM) [39] for the six available BCs presented in Table III. During model building, different speckle filters were tested independently, being only available six BCs each time. The ten available texture features in SNAP (presented in Table III) were obtained with a 5×5 window, with displacement of 1, in all the directions with 64 quantization levels. The second feature set consisted of three polarization ratios and normalized differences. Equations (2) and (3) were then applied to L-band gamma nought, and C-band gamma and sigma nought values (computed using SNAP). Finally, the principal component analysis was applied to the BCs extracted from ALOS-2 and S1, independently, giving two and four principal components, respectively, using

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TABLE II
SUMMARY OF THE USED DATA PER STUDY AREA

Study Area ID	Study Period	ALS Campaign Dates (YYYYMMDD)	LiDAR Sensors	Point Clouds Resolution (points/m^2)	ALOS-2 Acquisition Date	ALOS-2 Tiles	Number of S1 Observations	LCLU Map
1	2020	20200619	Teledyne Galaxy PRIME Airborne Lidar Terrain Mapper	10.02 – 13.88	20200523	N39W010	31	COS 2018
2	2020	20200614			20200523	N39W010	31	
3	2020	20200619			20200425	N40W009	30	
4	2020	20200621			20200504 20200518	N41W009	30	
5	2020	20200614			20200523	N39W010	31	
6	2020	20200620			20200420 20200518	N42W008	29	
7	2020	20200713 20200823 20200825 20200826	Optech Prime, ALS80HP	1.50	20200513 20200527	N37W006	31	CORINE 2018
8	2020	20201008 20201102	Optech Prime	1.50	20200423	N38W003	30	
9	2020	20200802 20200803 20200814	RIEGL VQ-1560i	2.00	20200703	N40W005	29	
10	2020	20201030 20201031 20201101	Optech Prime	1,50	20200712	N39W003	30	
11	2020	20210206 20210217 20210418	Leica ALS80	1.50	20200527	N38W006	31	
12	2021	20210519 20210520	Optech Prime	1.00	20210426 20210505	N43W001	31	
13	2021	20210627 20210701 20210703	Optech Prime	1.00	20210510	N41W002	31	
14	2020	20201027 20201029 20201103 20210218 20210219	Optech Prime, Optech T2000, ALS80HP	1.50	20200522	N38W004	30	
15	2020	20200929 20201008 20201009 20201010 20201012	Optech Prime	1.50	20200424 20200522	N37W003 N37W004 N38W003 N38W004	30	
16	2019	20190616	ALTM Gemini LiDAR	2.00 – 6.00	20191019	N37W120 N37W120	31	NLCD 2019

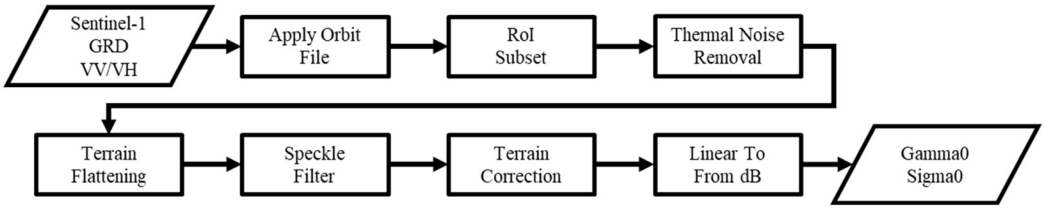


Fig. 3. S1 preprocessing diagram.

the scikit-learn (Python module)

$$\text{Polarization Ratio} = \frac{\text{BC}_{\text{Copolarized}}}{\text{BC}_{\text{Crosspolarized}}} \quad (2)$$

$$\text{ND} = \frac{\text{BC}_{\text{Crosspolarized}} - \text{BC}_{\text{Copolarized}}}{\text{BC}_{\text{Crosspolarized}} + \text{BC}_{\text{Copolarized}}}. \quad (3)$$

C. Temporal Compositing

The computation of the temporal composite was the first stage of features' processing. It was only applied to S1-related features, since for ALOS-2 there is only one observation for the defined study periods. This six-month temporal compositing stage was included in the model to decrease the variability

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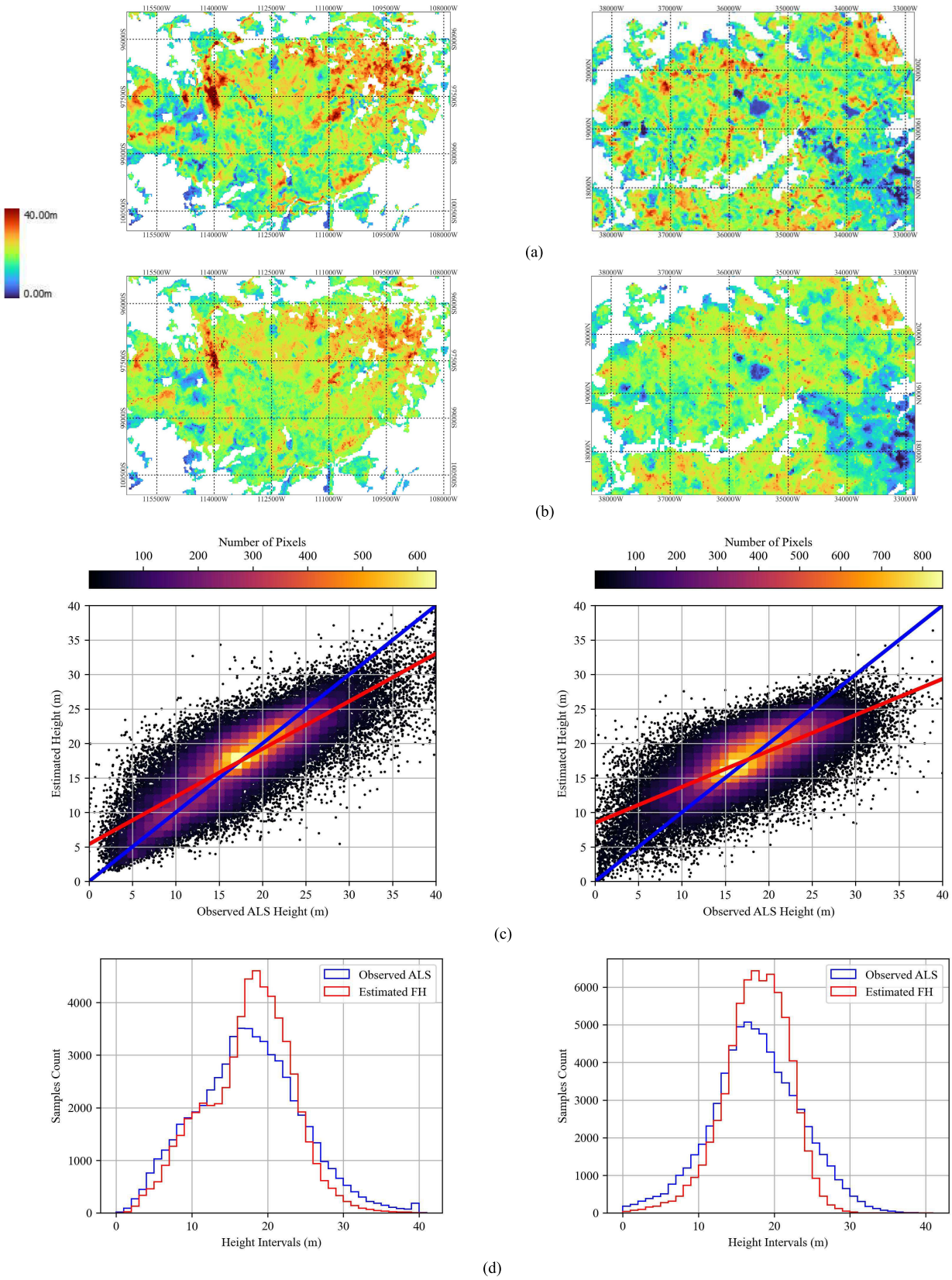


Fig. 4. Results for study areas (right) 5 and (left) 3. (a) CHM from the observed ALS. (b) Estimated FH map. (c) Observed ALS versus estimated height scatter plot (with a 1:1 line in red and linear trend line in blue). (d) Histograms. (a) and (b) Area inside the study area using EPSG: 3763.

of the observations. Two approaches were compared: mean and median, being calculated for each pixel using the six-month observation period.

D. Spatial Filtering

The spatial filtering was used to decrease data spatial noise. Two filters were tested: the median and mean filter, both

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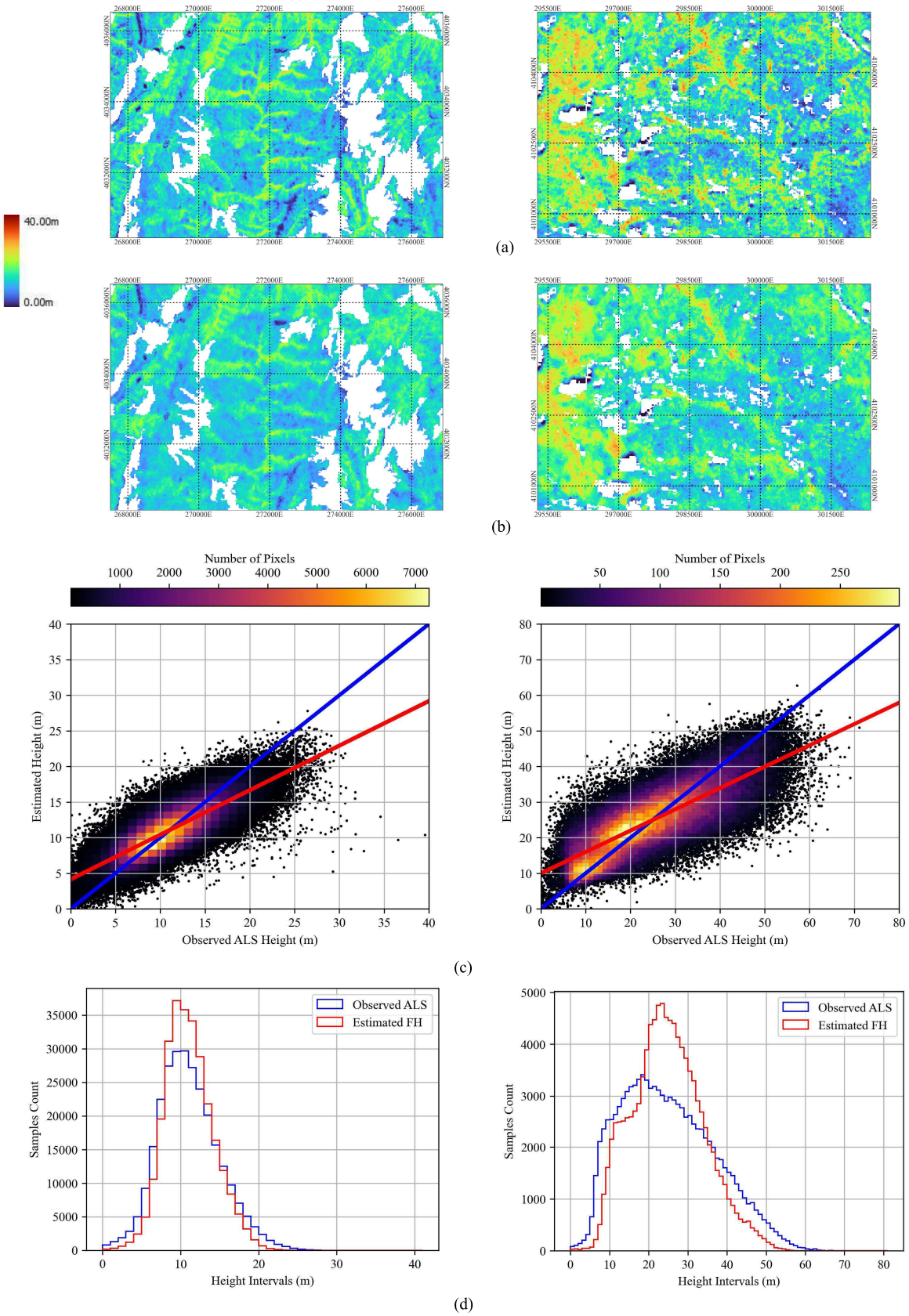


Fig. 5. Results for study areas (left) 7 and (right) 16. (a) CHM from the Observed ALS. (b) Estimated FH map. (c) Observed ALS versus estimated height scatter plot (with a 1:1 line in red and linear trend line in blue). (d) Histograms. (a) and (b) Area inside the study area using (left) EPSG: 25830 and (right) EPSG: 32611.

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TABLE III
SUMMARY OF THE RM FEATURES

Feature Set	Features
Backscatter Coefficients	ALOS-2: Gamma0_HH, Gamma0 HV S1: Gamma0_VV, Gamm0_VH, Sigma0_VV, Sigma0_VH
Textural	Contrast, Dissimilarity, Homogeneity, Angular Second Moment, Energy, Maximum Probability, Entropy, GLCM Mean, GLCM Variance, GLCM Correlation
Ratios & normalised differences	ALOS-2: Gamma0 Ratio and normalised difference S1: Gamma0 and Sigma0 Ratios and normalised differences
Principal Components	ALOS-2: PC1, PC2 S1: PC1, PC2, PC3, PC4

with 3 × 3 kernel. If these filters were indiscriminately applied, at the forest border pixels would be included in the median/mean calculation. To prevent this from happening, nonforest pixels within its kernel were replaced by the closest forest pixels' values. Another source of nonforest effects in the data was the fact that the border pixels themselves might cover forest and nonforest areas. This was a consequence of the resampling applied to the satellite imagery and LCLU maps, and their different spatial resolutions. The spatial filtering was conducted using the ndimage module from Scipy library.

E. FH Mapping

Two steps compose FH mapping of the study area: local calibration and FH prediction. For local calibration, the study area was divided into a grid of cells with a spatial resolution of 25 m. Then, the calibration cells' location was randomly selected and had its FH measured. The calibration cells' FH measures were the ones acquired by ALS missions. Since one of the objectives was the use of small datasets for FH mapping, calibration datasets using 25%, 20%, 15%, 10%, and 5% of the available grid cells were tested, leading to the study areas' coverage presented in Table IV. The mapping was built by estimating the FH for the cells that were not included in the calibration dataset, using the locally calibrated machine learning regressor.

Three machine learning approaches were evaluated: random forest [40], gradient boosting [41], and stacking regressor [42] (all using scikit-learn implementations). The latter is an ensemble regressor that uses a layer structure. Each layer has a set of different regressors to estimate the target variable. These estimates are the features of the next layer, which produces new estimates. The same repeats until the last layer, which is composed by a single regressor computing the final estimate of the target variable. Here a stacking regressor with a two-layer structure was used, having on its first layer a random forest and a gradient boosting, and in the output layer a linear regression.

F. CHM Generation

The local calibration datasets used for FH mapping were obtained from the study areas' CHMs available in the Zenodo database [43]. These CHMs were produced at a 25-m spatial resolution, by applying the pit-free algorithm (from the lidR package [44]) to the normalized ALS point clouds (previously computed in LASTools).

G. Model Building and Testing

The first part of model building consisted of the comparison between different speckle filters, temporal composites, and spatial filter, being selected the approaches that led to the best mapping results. For this comparison, all the features were considered and a random forest (with 150 trees and a maximum depth of 30) was used for the regression of the FH. In the second part, the best feature sets were identified, and the same test and selection process was done for the machine learning regressors (and their hyperparameters). The model building evaluation process was as follows.

Random splitting of the FH reference values from the study area's CHM in a calibration and a validation set, with the proportions 25% and 75%, respectively.

Fitting of the model's regressor using the small calibration dataset.

Calculation of the coefficient of determination (R^2), root mean square error (RMSE), and relative RMSE (rRMSE), exclusively for the validation set.

During validation, where only the Portuguese study areas were considered, the process was repeated ten times per study area, and their average results were calculated. Note that since one of the objectives of this article was to define a model capable of mapping the FH with small calibration datasets, only 25% of the dataset was reserved for training. Finally, the standard deviation (SD) of the results obtained for the validation study areas was computed. During model building, the best methods and feature sets were selected based on the lowest rRMSE.

After defining the model, it was tested for the Spanish and Californian study areas using the previous process, but only performing it one time, to understand its applicability to new study areas.

V. RESULTS

The proposed FH mapping model achieved an average R^2/rRMSE of 0.60%/23.27% (being it for the Portuguese areas 0.62%/24.30% and for the remaining 0.59%/22.64%). When the speckle and spatiotemporal filtering stages were not included, the results achieved were 0.47%/28.85%. Note that during model building, only the Portuguese areas were included. This means that the inclusion of the proposed steps led to an improvement of 0.15/4.55 pp. Throughout this section, the results achieved during model building and testing will be presented, and the impacts of each stage and different methods are presented.

A. Preprocessing Stages

The results relative to speckle filtering are presented in Table V. The Lee filter with 7 × 7 kernel led to the best results achieving 0.53%/27.20%, which was better than not including a speckle filter by 0.06/1.65 pp. The use of DEMs derived from SRTM led to a higher accuracy (0.53%/27.20%)

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TABLE IV
NUMBER OF CELLS AND EXTENSION OF THE AREA (IN BOLD) USED FOR CALIBRATION FOR DIFFERENT STUDY AREAS AND DIFFERENT PROPORTIONS

Study Area ID	Proportions (%)											
	All		25		20		15		10		5	
	Points	Area (ha)	Points	Area (ha)	Points	Area (ha)	Points	Area (ha)	Points	Area (ha)	Points	Area (ha)
1	35652	**2228**	8913	**557**	7130	**446**	5347	**334**	3565	**223**	1782	**111**
2	13700	**856**	3425	**214**	2740	**171**	2055	**128**	1370	**86**	685	**43**
3	91776	**5736**	22794	**1425**	18235	**1140**	13676	**855**	9117	**570**	4558	**285**
4	249104	**15569**	62276	**3892**	35228	**2202**	37365	**2335**	24910	**1557**	12455	**778**
5	76688	**4793**	19172	**1198**	15337	**959**	11503	**719**	7668	**479**	3834	**240**
6	113764	**7110**	28441	**1778**	22753	**1422**	17064	**1067**	11376	**711**	5688	**356**
7	369324	**23083**	92331	**5771**								
8	245184	**15324**	61296	**3831**								
9	155948	**9747**	38987	**2437**								
10	351332	**21958**	87833	**5490**								
11	285368	**17836**	71342	**4459**								
12	229564	**14348**	57391	**3587**								
13	101836	**6365**	25459	**1591**								
14	103988	**6499**	25997	**1625**								
15	479244	**29953**	119811	**7488**								
16	138760	**8673**	34690	**2168**								

than that derived from AW3D30 (0.49%/28.26%). When RM only resorted to the S1 gamma nought features, it achieved an R^2/rRMSE of 0.17%/36.21%, while with the sigma nought features only the results were 0.17%/36.24%. With the use of both the backscattering feature sets, the same metrics improved to 0.24%/34.73% (i.e., improved by 0.07/1.48 pp).

The temporal composite stage was responsible for improving FH mapping by 0.02/0.77 pp (compared with when only speckle filtering was used), to R^2/rRMSE (SD) of 0.56%/26.43% (0.06%/3.38%). These results were achieved when a mean temporal composite was used, which revealed to be superior compared with the median. The results achieved by different spatial filtering approaches are presented in Table V. As can be seen, the best filter in this case was the median filter which led to 0.59%/25.19% with an SD of 0.06%/3.37% (improvement of 0.04/1.24 pp to the results achieved with speckle filtering and temporal composite). To summarize, a Lee speckle filter with 7 × 7 kernel, a six-month mean composite, and a corrected median spatial filter were selected for the preprocessing stages with the SRTM used as DEM. Despite the improvement generated from these stages, this model was still worse than the final model by 0.03/0.89 pp.

B. Impact of Multifrequency Datasets

The model, when using only L-band features, mapped the FH with an R^2 of 0.35 and rRMSE of 32.22%, while with C-band features these were 0.40% and 30.86% (see Table VI). Without spatiotemporal filtering, the results achieved by both the L-band and C-band feature sets were 0.30%/33.36% and 0.25%/34.52%, respectively. Finally, the use of multifrequency dataset improved the results to 0.42%/30.31%. This shows that the FH was better mapped by the proposed model when features from the multifrequency dataset were used. At first glance, C-band features seemed to be more suitable for FH mapping, since they achieved better results than L-band features by 0.05/3.67 pp. However, this was mainly due to temporal compositing. If no spatiotemporal filtering is applied to the datasets (i.e., the features are used directly for FH mapping), the L- and C-band features achieved 0.30%/33.36% and 0.25%/34.52%, respectively. This means that the L-band results were better by 0.05/1.16 pp than the ones for the C-band.

C. Features Selection

Table VI shows the results of the model for FH mapping when using different combinations of the feature sets presented in Table III. The textural features resulted the most important for FH mapping, achieving R^2/rRMSE of 0.51%/27.72%. When ancillary features were added, these values were improved by 0.08/2.52 pp. Furthermore, using all the features with the exception of the ancillary ones, the results were worse by 0.08/2.37 pp, than using just the texture

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TABLE V
SPECKLE FILTER RESULTS

Speckle Filter	Kernel	Sigma	R^2	R^2 SD	rRMSE (%)	rRMSE SD (%)
None	-	-	0.4704	0.0514	28.85	3.15
Lee	3x3	-	0.4871	0.0641	28.40	3.28
	5x5	-	0.5082	0.0648	27.84	3.34
	7x7	**-**	**0.5296**	**0.0639**	**27.20**	**3.29**
Lee Sigma	5x5	0.5	0.4998	0.0596	28.06	3.19
	5x5	0.7	0.5088	0.0575	27.79	3.19
	5x5	0.9	0.5098	0.0506	27.76	3.10
	7x7	0.5	0.5162	0.0668	27.58	3.56
	7x7	0.7	0.5212	0.0581	27.44	3.35
	7x7	0.9	0.5250	0.0536	27.32	3.22
Refined Lee	7x7	-	0.4968	0.0572	28.14	3.24

and ancillary sets, reinforcing the impact of this feature set. Therefore, the selected feature groups to be used by the model were: textural, ratios and normalized differences, principal components, and ancillary features.

D. FH Mapping

Different hyperparameters were evaluated both for random forest and gradient boosting. A random forest with 400 trees with a maximum depth of 30 and a gradient boosting with these hyperparameters set to 500 and 10, respectively, were used. Finally, for the stacking regressor, both the first layer regressors used the previously defined hyperparameters.

The mapping using the random forest achieved an R^2/rRMSE of 0.60%/25.02%, and for gradient boosting the results were 0.61%/24.74%. The combination of the previous machine learning regressors in the proposed stacking regressor was better than using the gradient boosting by 0.01/0.44 pp (Table VII presents the results per study area).

Table VIII presents the performance of the model using smaller calibration datasets. Note that the number of samples and area covered by these datasets are provided in Table IV. It can be highlighted that when it was reduced to 10% of the available grid cells, the model achieved 0.53%/27.19%, and the reduction to 5% led to an R^2 of 0.47 and an rRMSE of 28.87%.

TABLE VI
RESULTS OF USING DIFFERENT FEATURES' COMBINATIONS

Features Combination	R^2	rRMSE (%)
All	0.5958	25.19
L-band	0.3453	32.22
C-band	0.3990	30.86
Backscatter Coefficients	0.3065	33.22
Textural	0.5137	27.72
Ratios & normalised differences	0.1197	37.33
Principal Components	0.3204	32.85
Textural Ancillary	0.5961	25.20
Backscatter Coefficients Textural Ancillary	0.5952	25.22
Textural Principal Components Ancillary	0.5957	25.22
Textural Ratios & Normalised Differences Ancillary	0.5974	25.15
Backscatter Textural Ratios & Normalised Differences Principal Components	0.5193	27.57
Textural Principal Components Ratios & Normalised Differences Ancillary	**0.5983**	**25.11**

E. Final Results by Study Area

After the definition of the most suitable parameters for the model during the building stage, done in the study areas of Portugal, it was tested for the remaining study areas of California and Spain. The model could map FH in Mediterranean forests, resorting to local calibration datasets, with an R^2/rRMSE of 0.62%/24.30% for the validation study areas and 0.59%/22.64% for the testing study areas. In addition, the total improvement of the proposed model was 0.15/4.55 pp, when compared with using the features directly in a random forest. Moreover, the SD of the improvements achieved by all the study areas was 0.02 for R^2 and 0.73% for rRMSE, showing that the proposed methodology has a similar impact for all of them.

In Figs. 4 and 5, the resulting maps, histograms, and scatter plots for 3, 5, 7, and 16 are presented (which include study areas of all the countries and forest types here considered). The model building and test results showed that the former had a better R^2 by 0.03 and a worse rRMSE by 1.66 pp. This demonstrated the applicability of the proposed model to a wide set of Mediterranean study areas. Another fact is that the use of different LCLU maps depending on the study area did not

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TABLE VII
MODEL RESULTS PER STUDY AREA. *VALIDATION STUDY AREAS

Study Area ID	R^2	rRMSE (%)	RMSE (m)	MAE (m)
1*	0.5569	29.08	3.77	2.82
2*	0.6653	19.82	3.47	2.50
3*	0.5300	22.97	4.05	3.10
4*	0.6760	24.55	3.76	2.79
5*	0.6891	21.91	3.83	2.86
6*	0.6256	27.48	2.96	2.10
7	0.6310	21.62	2.40	1.82
8	0.5758	21.41	1.88	1.35
9	0.6040	27.70	2.39	1.61
10	0.5117	18.43	3.08	2.31
11	0.4528	20.32	2.18	1.54
12	0.5610	28.21	3.14	2.19
13	0.5968	14.90	2.03	1.51
14	0.7114	22.69	2.26	1.64
15	0.6494	21.45	2.24	1.61
16	0.6069	29.70	7.42	5.66
Validation Study Areas	**0.6240**	**24.30**	**3.64**	**2.68**
Test Study Areas	**0.5901**	**22.64**	**2.90**	**2.12**

TABLE VIII
RESULTS ON VALIDATION STUDY AREAS USING DIFFERENT CALIBRATION SET SIZES (THE VALUES INHA ARE PROVIDED IN TABLE IV)

Calibration Set Size (%)	R^2	rRMSE (%)	RMSE (m)	MAE (m)
25	0.6240	24.30	3.64	2.68
20	0.6011	25.04	3.75	2.77
15	0.5687	26.04	3.90	2.89
10	0.5293	27.19	4.08	3.04
5	0.4705	28.87	4.33	3.25

have a significant impact on FH mapping accuracy. Looking at the regions individually, the best results were achieved for Sierra Magina and the worst for Constantina. Furthermore, the coefficient of variation (CV) of the Sierra Magina dataset was 0.42, while the Constantina dataset was 0.27, which showed that the former is a more diverse dataset and 0.65 for the six mixed forests. In addition, the average R^2 was 0.56 for the four broad-leaved study areas, 0.60 for the five coniferous and 0.65. Fig. 5(b) shows the capacity of the model for accurately mapping the Californian study area and capturing its broader range of FH values.

VI. DISCUSSION

A. Model

The results achieved during model building show that all the proposed stages have a positive impact on FH mapping. Regarding the methods, the most adequate stages were the Lee filter (speckle filtering), mean temporal compositing, updated mean spatial filter, and stacking regression.

The fact that the redefined Lee (which is tailored for heterogeneous areas) was the least effective speckle filtering approach may indicate that speckle noise reduction at the forest border is less important than speckle noise reduction at homogenous areas, which is more effectively addressed by the two other filters.

Looking at spatial filtering, the median outperformed the mean. This was expected due to its improved ability to ignore outliers compared with the mean filter. In addition, the proposed use of LCLU in spatial filtering led to improvements in all the study areas. Therefore, it might be said that it decreased the nonforest effects, being the final pixel value computed using only the surrounding forest values. This approach also enabled the use of all pixels, instead of creating a pixel buffer at the borders, as was done in [7].

The advantages of the synergy between L- and C-band sensors were demonstrated in this article. This can be explained by the fact that the sensors operate in different regions of the spectrum and, due to their different penetration depths, provide different backscattering mechanisms (L-band has more double-bouncing and C-band more volumetric backscattering). This further highlights the importance of NISAR, which will start providing L-band data more frequently than ALOS-2. This will allow temporal compositing of the L-band, which revealed to be significantly beneficial for the C-band. Looking at the two available BCs from S1, the sigma and gamma nought, better results were achieved when both were used jointly.

The features' evaluation showed that when different sets were used alone, textural features achieve the best results. This was expected, since the textural set was composed of 60 of the total 82 features assessed here. The analysis done shows that the inclusion of BCs was not beneficial, which can be explained by the fact that the backscatter information was already included in other feature sets, so their information was redundant. Regarding machine learning regressor, stacking regression proved that the combination of using a random forest or a gradient boosting by themselves would be worse.

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TABLE IX
IMPORTANT RESULTS FOUND IN THE LITERATURE

	Platform/ Data	Region	Forest Type	Calibration Test Size	Resolution	Approach	RMSE (m)
[46]	TanDEM-X	Mexico, Mozambique	Mangrove	-	12m	RVoG	1.27-1.39
[45]	TanDEM-X, TerraSAR-X	Germany, Sweden, Indonesia	Tropical, Temperate, Boreal	-	10m	RVoG	1.58-3.30
[18]	TanDEM-X, PNOA	Spain	Mediterranean	"Small Patch"	1m	C-SINC	1.50-3.42
[47]	TanDEM-X, Reigl & Leica	Germany, Canada	Temperate, Boreal	-	12m	SINC	4.10-5.30
[48]	Onera Sethi	French Guiana	Tropical	-	12m	TomoSAR	2.00-2.50
[22]	S1	Austria	Temperate	80%	100m	RF	4.76
[49]	Optech, S1, Landsat	Brazil, Panamá	Tropical	80%	30m	RF	1.22-5.13
[42]	ICESat-2, Sentinel-2	China	Temperate	70%	17m	SR	1.96
[9]	NAIP	San Francisco	Mediterranean	56%	1m	CNN	3.14
[50]	ICESat, Modis	Global	-	100%	500m	RF	2.75-4.45
[52]	GEDI, Sentinel-2	Global	-	80%	10m	CNN	6.00
[5]	GEDI, Landsat	Global	-	90%	30m	Bagged Regression Tree	6.60
	S1, ALOS-2	**Portugal, Spain, USA**	**Mediterranean**	**25%**	**25m**	**Stacking Regressor**	**1.88-7.42**

Looking at the SD values achieved during model building, they remained almost unchanged when new stages were included in the model, showing that they add a similar impact on all study areas.

The two examples provided in Fig. 4, obtained during validation, show that despite accurate FH mapping, the model had some difficulties in estimating the underrepresented values. This can be seen by the reduced number of high FH (i.e., areas) red areas in the produced maps in both the cases. In addition, their histograms demonstrate the tendency of the model to concentrate its estimates on the FH most represented in the dataset. The main reason behind this is the fact that the available datasets are imbalanced. This means that there are FH values substantially more represented in the calibration datasets than others, leading to the final model being more accurate in predicting more represented values, and less accurate for others. A consequence of this is the under- and overestimation of FH, at higher and lower values, respectively.

Despite the positive impact that the spatial filter brought, it smoothed the calibration datasets, further reducing the sensitivity of the model. Notwithstanding this aspect of the model, both the histograms demonstrate that it has good sensitivity to the range of FH values of the study areas.

Finally, looking at the research questions defined in Section I-B, it can be said that the model developed in this work produces reliable maps using small calibration datasets, covering less than 25% of the study area. In addition, the use of multifrequency SAR data had a positive impact on mapping.

B. Comparison to the Literature

Table IX presents the RMSE (as it was the common metric found in all of them) of 12 relevant studies found in the literature that map FH using remote sensing data. Here the achieved RMSE ranged from 1.88 to 7.42 m, and if Soaproot Saddle is not included (which has FH values of up to 80 m) the maximum RMSE was 4.05 m. Gomez et al. [18] mapped the FH for four Spanish study areas with a C-SINC using data from TanDEM-X and PNOA and achieved an RMSE of 1.50–3.42 m. Still on Mediterranean forests, in [9] the use of a CNN to estimate the FH in San Francisco's study area led to an RMSE of 3.14 m. Data from a multispectral sensor were used, and the CNN was calibrated using 56% of the study area data. In [22], FH was mapped with a random forest and S1 observations, achieving an RMSE of 4.76 m, in an Austrian region, while using 80% of data for calibration.

In [45] and [46], the RVoG model was used to map the FH in tropical, temperate, and boreal forests, achieving an RMSE in between 1.27 and 1.39 m and 1.58 and 3.30 m, respectively, using data from TanDEM-X. In [47], a simpler approach, the SINC model, was used, achieving an RMSE of 4.10–5.30 m. Alternatively, in [48] TomoSAR techniques

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were used, leading to an RMSE of 2.00–2.50 m in a study area in French Guiana. In [42], a stacking regressor was used to map the FH in a study area in China, leading to an RMSE of 1.96 m, using 70% of the data for calibration. Fagua et al. [49] tested the combination between SAR and multispectral sensors (S1 and Landsat) that map three study areas across Brazil and Panamá with RMSEs between 1.22 and 5.13 m, while using a random forest. Finally, Potapov et al. [5], Wang et al. [50], and Lang et al. [51] mapped the FH globally. In [50], a random forest was used with ICESat and MODIS data reaching an RMSE of 2.75–4.45 m for seven test study areas, calibrating with all the available data.

Lang et al. [51] achieved an RMSE of 6.00 m using a CNN to estimate the FH with data from GEDI and Sentinel-2. Note that mapping was done per tile, using 80% of the available data for calibration. In [5], a bagged regression tree with GEDI and Landsat achieves an RMSE of 6.60 m while using 90% of the available data for calibration.

The results presented in Table IX for Mediterranean forests were in between 1.50 and 3.42 m for a total of four study areas. This article proposed a model that achieved comparable results for 16 study areas. Looking at different approaches, the use of coherence-based models had the advantage of not requiring calibration data (with the exception of C-SINC); however, they did not use free access data, which makes them less suitable for large-scale mapping and operational scenarios. In the study here proposed, the RMSE ranges from 1.27 to 5.30 m. For the six articles that proposed the use of machine learning regressors, as in here, the RMSE is 1.22–6.00 m, while using calibration sets with 56%–100% of the available data. The results in line with these were achieved with the model here proposed, which only required 25% of the available data.

The three studies dedicated to map the FH globally (large-scale mapping) had an RMSE between 2.75 and 6.60 m with the advantage that relied on spaceborne LiDAR data for calibration, being independent of ALS and field campaigns. Their methodologies resorted to multispectral data, instead of SAR data as here. However, it is important to consider the following two comments. The first is that machine learning models fit at global scales produce biased estimates of heights when used at regional or local scales [5], [50], [52].

The first is that machine learning models fit at global scales produce biased estimates of heights when used at regional or local scales [5], [50], [52]. The second is that most studies randomly select 10%–20% of the CHM dataset as independent test dataset (which was not used for training) and to optimize the model parameters during training. However, FH estimation using machine learning regressors for more limited local calibration datasets had not been explored in the literature. To the best of our knowledge, this is the first study to investigate the capacity of fitting local FH regressor testing 25%, 20%, 15%, 10%, and 5% of the available data for training and a robust validation for testing. The study proposes a novel method to obtain accurate local mapping height, which is crucial for making informed decisions that balance the ecological, economic, and social aspects of forest management. The proposed method could avoid the limitations of FH global or regional maps for capturing detailed variations in FH that are essential for precise management decisions at stand level.

The results achieved were in line with the literature, while using substantially smaller calibration datasets (comparing with the machine learning approaches) or freely available data (comparing with the coherence-based approaches). Despite these advantages, there are two main drawbacks. The fact that it still requires ALS data for calibration, in opposition to the coherence-based approaches and global mapping methodologies, and the coarser resolution when the literature here presented is considered (eight references have resolutions between 1 and 17 m).

VII. Conclusion

The model proposed in this study proved to be capable of accurately mapping the FH at different Mediterranean study areas using multifrequency SAR data and ancillary features, calibrated with small datasets. Using both the BCs of S1, instead of just one of them, was beneficial. The comparison between the L- and C-band for FH mapping showed that the first achieved better results. However, when a temporal composite was applied to the C-band (which was not possible for the L-band), the C-band was better, which further highlighted the importance of temporal composite.

It was demonstrated that the inclusion of LCLU in filtering further improves FH mapping. The fact that the SDs were nearly the same whenever a new stage was included in the model showed that they impacted all the study areas in the same way.

The use of stacking regressor that combined a random forest and a gradient boosting using a linear regression proved to be better than using these regressors individually. The synergy between both the satellites significantly benefits the estimation, showing the positive impact of using multifrequency datasets.

But, most importantly, the use of a diverse set of regression features strongly improved FH mapping (namely, the use of the texture and ancillary features). Therefore, all the stages included in the proposed model allowed mapping of the FH with an average rRMSE below 30%, while using calibration datasets that only cover 5%–25% of the study area.

Another interesting conclusion that can be drawn from this article is that the redefined Lee revealed to be the less effective speckle filter for FH mapping. The other two filters here tested achieved similar results. Comparing to the single-pass InSAR and TomoSAR approaches, the proposed model has the disadvantage of requiring calibration data. However, it has the advantage of only using freely available data.

Several steps are being considered for further development of the work here presented. First, the use of multispectral features will be compared with the results achieved here, and also the impact of combining both the SAR and multispectral datasets for FH mapping. Second, the use of different temporal composites considering leaf-on and leaf-off conditions. Third, the use over- and undersampling techniques is going to be tested in the attempt to address the problem of underrepresented FH values. In addition, the replacement of yearly ALOS-2 observation by the periodic L-band observations from the future NISAR mission is expected to bring further

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improvements. However, most regions of the world do not have ALS coverage to allow a direct application of the FH mapping method proposed in this article. Therefore, it will be essential to replace the ALS calibration datasets with datasets acquired by spaceborne LiDAR missions such as GEDI and ICESat-2. In addition, since these data sources will provide even smaller and sparser coverages than the ones used here, the model will need to be further improved to use even smaller calibration datasets.

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